

Advanced (Habanero)

Q1. Simplify $2^3 \cdot 2^4$

- (A) 2^{3+4}
- (B) 2^{3-4}
- (C) $2^{3 \cdot 4}$
- (D) $2^{3/4}$
- (E) $2^{4/3}$

Q2. Simplify $(3^2)^4$

- (A) 3^{2+4}
- (B) 3^{2-4}
- (C) $3^{2 \cdot 4}$
- (D) $3^{2/4}$
- (E) $3^{4/2}$

Q3. Simplify $\frac{4^3}{4^2}$

- (A) 4^{3+2}
- (B) 4^{3-2}
- (C) $4^{3 \cdot 2}$
- (D) $4^{3/2}$
- (E) $4^{2/3}$

Q4. Simplify $(2^3)^2$

- (A) 2^{3+2}
- (B) 2^{3-2}
- (C) $2^{3 \cdot 2}$
- (D) $2^{3/2}$
- (E) $2^{2/3}$

Q5. Simplify $\frac{5^4}{5^3}$

- (A) 5^{4+3}
- (B) 5^{4-3}
- (C) $5^{4 \cdot 3}$
- (D) $5^{4/3}$
- (E) $5^{3/4}$

Q6. Simplify $3^2 \cdot 3^5$

- (A) 3^{2+5}
- (B) 3^{2-5}
- (C) $3^{2 \cdot 5}$
- (D) $3^{2/5}$
- (E) $3^{5/2}$

Q7. Simplify $(4^2)^3$

- (A) 4^{2+3}
- (B) 4^{2-3}
- (C) $4^{2 \cdot 3}$
- (D) $4^{2/3}$
- (E) $4^{3/2}$

Q8. Simplify $\frac{2^5}{2^2}$

- (A) 2^{5+2}
- (B) 2^{5-2}
- (C) $2^{5 \cdot 2}$
- (D) $2^{5/2}$
- (E) $2^{2/5}$

Open Ended Questions (FRQ)

Q9. Simplify $\frac{(3^2)^3}{(3^3)^2}$

Q10. Simplify $\frac{2^4 \cdot 2^5}{2^3}$

Chef's Tips

- Tip 1: Check for any common factors in the numerator and denominator
- Tip 2: Use the product rule for exponents when multiplying like bases
- Tip 3: Use the quotient rule for exponents when dividing like bases